

4G DATA LOGGER WITH BATTERY

Hawk

The battery powered 4G logger
built for Australian sewer networks.

No mains, no cabinet

Internal battery up to 152 Ah and a 40 μ A sleep current.

Sealed for surcharge

IP68 canister and connectors; it keeps logging underwater.

Set up from the surface

Magnet wake and EMS-Flow set up; managed remotely over 4G.



**Logging from 1 second.
Reporting from 1 minute.**

4G LTE · 3 × RS485 · IP68 · up to 152 Ah

DESIGNED AROUND THE CHAMBER

Designed for the conditions inside a sewer maintenance hole.

A sewer maintenance hole is humid and corrosive, it can surcharge in wet weather, it has no power supply, and every entry is a confined space entry. The Hawk is designed to install and operate in that environment for long periods with minimal attention.



SURCHARGE

Sealed to IP68

Canister, lid and connectors are sealed for submersion, so the record continues through the event.



CORROSIVE ATMOSPHERE

Corrosion resistant

ABS canister, sealed connectors and a stainless wall bracket, chosen for the sewer atmosphere.



NO POWER AT SITE

Internal battery

72 to 152 Ah on board. No mains, no solar panel, no roadside cabinet.



CONFINED SPACE

Set up from the surface

Magnetic wake and EMS-Flow set up from the road; configuration and firmware follow over 4G.



LONG CABLE RUNS

15 kV ESD protection

Every RS485 port is protected against transients picked up on long sensor cables.



PATCHY COVERAGE

Every reading kept

32 MB on board holds 10 years of data, so a missed report does not mean lost data.

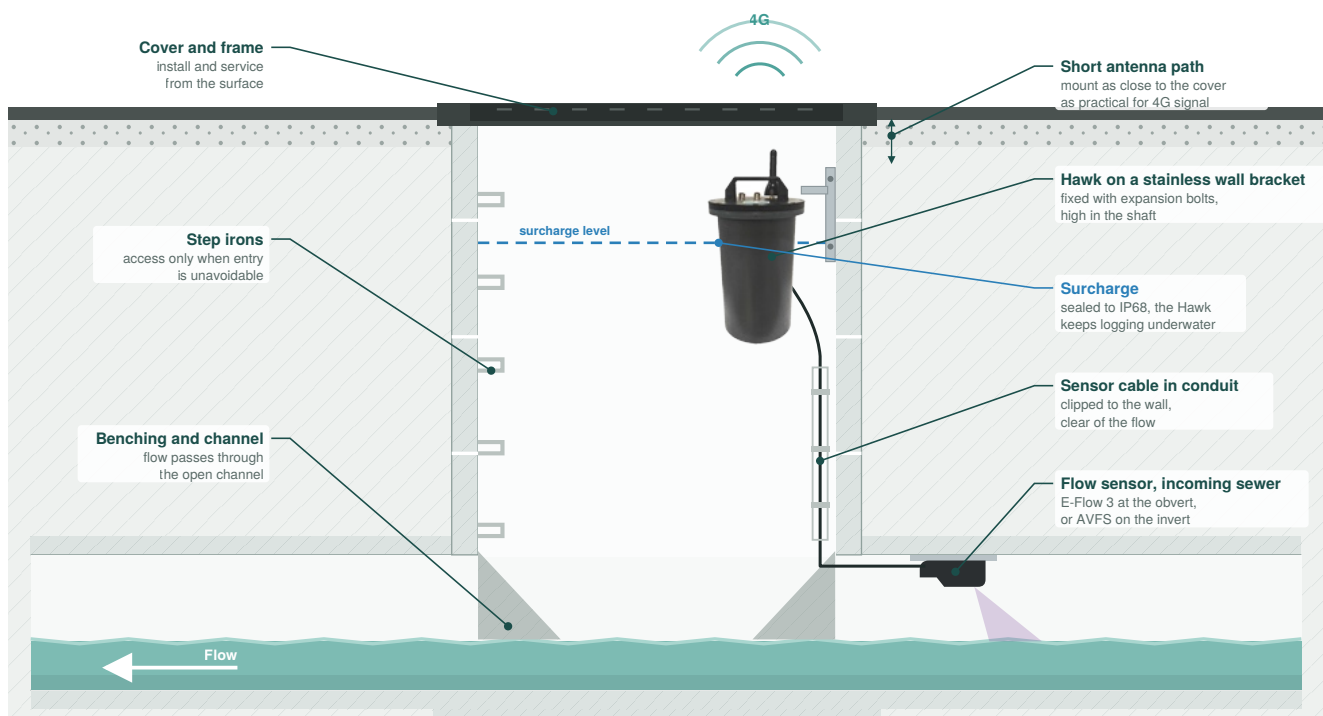


Supplied with a stainless wall bracket, fixed to the shaft wall with expansion bolts

INSTALLS FROM THE SURFACE

Installed below the cover and commissioned from the surface.

The Hawk mounts high in the shaft on a stainless bracket, close to the cover for the strongest 4G signal. Its sensor sits in the incoming sewer, with the cable in conduit down the wall. Nothing is mounted at the surface, so there is nothing to damage, vandalise or maintain in the road reserve.



Schematic, not to scale. Treat every maintenance hole as a confined space.

Figure 1. A typical installation. Schematic, not to scale.

1

Fit and seal

SIM fitted and the lid bolted down on its O ring seal.

2

Mount and connect

Bracket below the cover, sensors on RS485.

3

Wake and configure

A magnet wakes it; EMS-Flow sets it up over Bluetooth.

4

Confirm and leave

Confirm the first report in FlowSense, then replace the cover.

No confined space entry is needed to commission, reconfigure or update the Hawk.

Location

On the shaft wall clear of the step irons and cover opening, antenna up, as high as the cover and frame allow.

Fixing

Stainless wall bracket with expansion bolts. The canister hangs below the bracket and lifts out by its handle.

Safety

Treat every maintenance hole as a confined space and confirm the site hazardous area classification first.

SENSORS, LOGGER, TELEMETRY, DATA

Connects your sensors to FlowSense and to your own systems.

The Hawk powers each sensor only while it reads it, logs as often as every second, and reports over 4G LTE as often as every minute or as seldom as once a day. It can send the same data to FlowSense and to your own SCADA or historian at the same time, or run them as primary and standby, so the record continues through a platform outage.

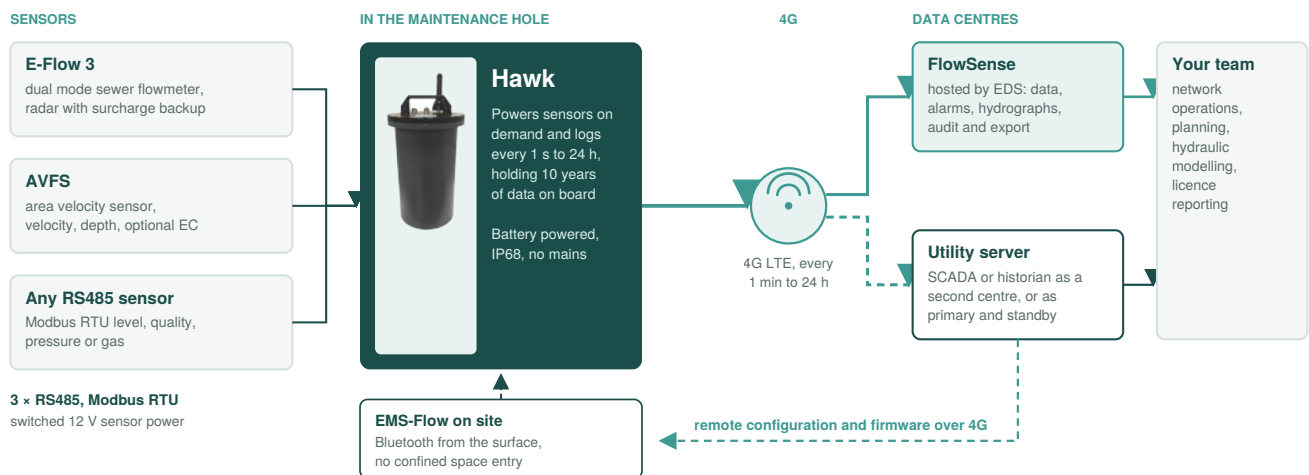


Figure 2. Up to three sensors on RS485, up to four data centres over 4G.

Works with EDS sensors

E-Flow 3 for radar flow with surcharge backup, AVFS for area velocity and conductivity, or any Modbus RTU sensor.

Two destinations at once

FlowSense for hosted data and alarms, and your utility server for SCADA and historian records.

Managed from the office

Change logging and reporting intervals in EMS-Flow, add a sensor channel or update firmware over 4G, without a site visit.

Flow surveys

Temporary gauging for hydraulic model calibration and upgrade planning.

Inflow and infiltration

Wet weather response across a catchment against its dry weather baseline.

Overflow structures

Level and spill detection at overflow and relief structures for licence reporting.

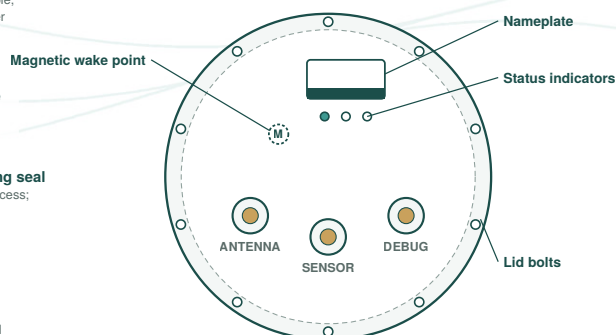
ANATOMY AND SPECIFICATIONS

Specifications at a glance

A THE HAWK



B TOP PLATE, FROM ABOVE



No switches or buttons: wake with a magnet, configure in EMS-Flow.

Top plate layout indicative.

TELEMETRY AND SENSORS

Cellular	4G LTE
Logging	Every 1 second to 24 hours
Reporting	Every 1 minute to 24 hours
Data centres	Up to 4, primary and standby
Sensor ports	3 × RS485, 15 kV ESD
Sensor power	2 × switched 12 V, 500 mA each
Configuration	EMS-Flow, on site or over 4G
Storage	32 MB, 10 years of data

POWER AND ENCLOSURE

Battery	72, 80, 100 Ah rechargeable or 152 Ah primary
Sleep current	40 µA or less at 7.2 VDC
Protection	IP68, V0 flame retardant ABS
Dimensions	Ø165 × 243 mm
Operating range	–40 to 60 °C, 95 % RH
Options	Methane, temperature and humidity, tilt

MODEL	BATTERY	BEST SUITED TO
Hawk 72	72 Ah rechargeable	Short surveys and sites visited regularly
Hawk 80	80 Ah rechargeable	General network monitoring
Hawk 100	100 Ah rechargeable	Long intervals and several sensors on one logger
Hawk 152	152 Ah primary	The longest unattended service, permanent and remote sites

Typical values. Battery life depends on logging and reporting intervals, connected sensors and site temperature; EDS sizes the battery for each site. Full specifications are in the Hawk datasheet, HAWK DS 001.

TALK TO EDS ABOUT THE HAWK

Reliable data from inside the sewer network, in dry weather and wet.

Tell us the network, the sites and the data you need. EDS will size the battery, supply the bracket and sensors, install and commission from the surface, and deliver the data through FlowSense. Hawk is also available with EDS rental fleets for temporary flow surveys.

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Environmental Data Services

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Australian owned and working in water and wastewater since 1991. Supply, installation, commissioning and support from Brisbane, with field crews across Australia.



Hawk on its stainless wall bracket